AGE INVARIANT FACE RECOGNITION AND RETRIEVAL FOR VIDEO ANALYTICS

21CN380 - DISSERTATION PHASE I

Submitted in partial fulfillment for the requirement of M.E. degree in Communication System of Anna University

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BONAFIDE CERTIFICATE

This is to certify that the **21CN380 - DISSERTATION PHASE I** entitled **"AGE INVARIANT FACE RECOGNITION AND RETRIEVAL FOR VIDEO ANALYTICS"**, being submitted by **ATSARA. P (Register Number 917721N002)** in partial fulfillment for the requirement of Masters of Engineering Degree in Communication Systems, is a record of Bonafide work done by her during the year 2022-2023 under my supervision. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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LIST OF ABBREVIATIONS

CNN	Convolution Neural Network
GAN	Generative Adversarial Network
RLN	Disentangled Representation Learning
FSS	Face Synthesis Subnet
AI	Artificial Intelligence
AIFR	Age-Invariant Face Recognition

ABSTRACT

The age recognition system is a technology which is proficient to match a human face from a video or from the database of digital faces. Age invariant Facial recognition systems has been positioned in law enforcement to identify the unknown individuals and determine their age at the time of death, health care who doesn't have correct birth records, video surveillance, etc. The human face is an intricate object and the various changes that it undergoes across ages is a long-winded process. The Age invariant face recognition and retrieval still remains a challenge. To resolve this problem an Age Invariant Face Recognition and Retrieval model (AIFRR) is proposed which comprises of three networks –Auto encoder based Generative Adversarial Network (GAN), disentangled Representation Learning sub-Net (RLN) and a Face Synthesis sub-Net (FSN) with the novelties which performs cross-age face synthesis and recognition. The proposed model achieves continuous face rejuvenation or aging with photorealistic and identity-preserving properties which will be carried out by the face synthesis subnet and mainly a new Age Invariant Face Recognition (AIFR) dataset is created which contains common people's face images at different age spans mostly based on the Indian people faces. Both gender and racial diversity considered where the data are organized to 10 splits. The results were categorized to four cases – best matched cases, best non matched cases, worst matched cases, worst non matched cases. Comparison results were taken on MORPH Benchmark data set and the obtained recognition rate is 9.13%. The face recognition performance comparison of the proposed AIFRR with other state-of-the-arts on labeled Faces in the Wild (LFW) is 0.998 accuracy percentage.

CHAPTER 1
INTRODUCTION

1.1 Facial Age Recognition and Retrieval

Facial age recognition and retrieval plays an important role in a variety of industries and applications. It is a technology which uses computer algorithms to estimate a person's age based on their facial features. The facial features are considered as the foremost personal characteristics which will differ slowly across ages. The roles of Age-invariant face recognition and retrieval (AIFRR) involves many practical applications. For in law enforcement scenarios - to identify suspects or missing persons and to assist in investigations, identifying wanted fugitives, marketing and advertising - to target ads and promotions which covers specific age groups, in order to reach their desired audience and in verifying passports that usually involves recognizing faces across ages. Since Facial characteristics changes largely over time with different age pattern per person. Extracting robust features describing aging facial details is challenging research problem.

By decomposing the facial representation into an age-specific component, it is possible to separate the aging effect from other factors such as facial structure, lighting, and expression. This can improve the accuracy of facial recognition systems, and to recognize a person across different ages. This provoked to develop AIFRR.

Age invariant face recognition and retrieval method identifies and retrieves the individuals based on their facial features, regardless of changes in their appearance due to aging. This can be accomplished by training a model to extract features from a face that are invariant to age. One approach for age-invariant face recognition and retrieval is to use deep learning techniques such as convolutional neural networks (CNNs) and autoencoders. These models can be trained on large datasets of faces labeled with age information.

1.2 Video Analytics

Video analytics technology is used to extract useful information from video footage, such as identifying patterns, detecting anomalies, and tracking objects or people. Video analytics can be applied to both live and recorded videos. The main goal of video analytics is to automatically recognize temporal and spatial events in videos. Video analytics use deep learning for video classifications where a model is trained to classify the content of a video into different categories. For processing an image video analytics

use deep learning techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs).

Video analytics in age detection refers to the use of computer vision and machine learning techniques to automatically detect and estimate the age of individuals in video footage. This can be done by analyzing facial features. This technology can be used in a variety of applications, such as security and surveillance, demographic analysis, and targeted advertising. The main step of video analytics is to analyze the video data. This can include tasks such as object detection, face identification This step is typically performed by a software that is specifically designed for video analytics. Figure 1 represents process involved face recognition using video analytics.

Figure 1: Video analytics in face recognition

1.3 Deep Learning

Deep learning is a touch of machine learning which involves training artificial neural networks for large amounts of data. Both Machine learning and deep learning comes under the artificial intelligence (AI) that is inspired by the structure and function of the human brain. Deep learning models are designed by the reference of the human brain and they are composed of layers of interconnected neurons. Each and every layer process the input and passes it to the next layer. The layers are connected by pathways called

synapses. Deep learning models have some specialized layers, such as convolutional layers - processing image data and recurrent layers - processing sequential data.

Deep learning has proven to be a very effective way for age and face detection, with accuracy rates which are significantly higher than the traditional methods. There are several methods for age detection using deep learning.

- Multi-task learning
- Regression-based
- Classification-based
- Adversarial-based

Multi-task learning - This approach trains a model to perform multiple tasks, such as age detection and facial recognition. The model learns to extract features which will be used for both tasks, and the age detection performance is improved by the auxiliary task of facial recognition.

Regression-based - This approach trains a on a dataset of images with labeled ages, and based on the trained data model to predict the age of an individual based on their facial features. The model is trained set the model learns to recognize patterns in the facial features that are unified to different with each age groups.

Classification-based - This approach involves training a model to classify a person's age into predefined age groups, such as 0-10, 11-20, etc. As similar to the regression based method this method also recognizes the facial feature for each age group.

Adversarial based – This approach has two models, generative model and discriminative model. Firstly the generator creates the fake images of faces with different ages and a second model, tries to distinguish between the fake and real images.

1.3.1 Adversarial Based Deep Learning

Adversarial based deep learning is a technique that trains a model by using adversarial samples, whose inputs are wantedly mislead to fake the discriminator. Adversarial based deep learning is the intense method of producing high-quality and realistic results. Yet it can also be challenging to train the whole

model, as the generator and discriminator can sometimes get annexed. In addition to this GANs are known to be difficult to train, and the stability of the training process is still an active area of research.

1.3.1.1 Generative Adversarial Networks

Generative Adversarial Networks (GAN) are training data. Figure 2 represents the architecture of GAN. The main goal of the generator is to fool the discriminator, which is another neural network that is trained to distinguish real samples from the generated samples. Both generator and discriminator are trained together in an adversarial manner, with the aim of generator produced samples are not distinguished real samples, and the discriminator trying to correctly classify the fake and real samples.

The discriminator in a GAN acts as a guide for the generator, helping it by pointing out the errors. Discriminator act as a critic for the generator by distinguishing the real samples from the generated ones and providing feedback to the generator to improve the ability to generate more realistic samples.

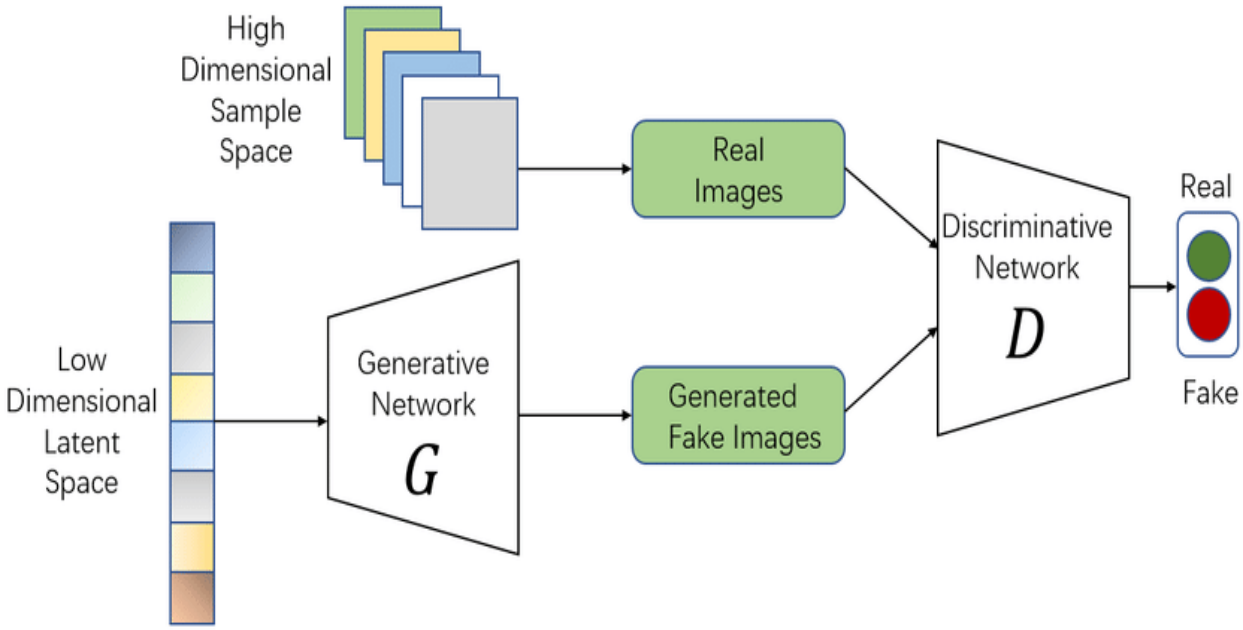

Figure 2: The Architecture of Generative Adversarial Network

1.3.1.2 Autoencoders

Autoencoders are neural networks that can be used for image processing tasks such as image compression and denoising. The design of an autoencoder is shown in Figure 3. An autoencoder is composed of two main parts: an encoder and a decoder. The encoder takes in an image as input and compresses it into a lower dimensional representation, called the bottleneck or latent representation. The decoder takes this bottleneck representation and produces an output that should be similar to the original input image.

In image compression, the encoder is trained to learn a compact representation of the image, while the decoder is trained to reconstruct the original image from this compact representation. The encoder can be used to compress the image by reducing the number of bits needed to represent it.

In denoising, autoencoders are trained to reconstruct the original image from a noisy version of it, by learning to remove the noise from the image. The encoder and the decoder will learn to extract the important features from the image and discard the noise.

Additionally, autoencoders can be used for image generation, where the encoder learns to map the image to a low-dimensional representation, and the decoder generates new images from this low-dimensional representation.

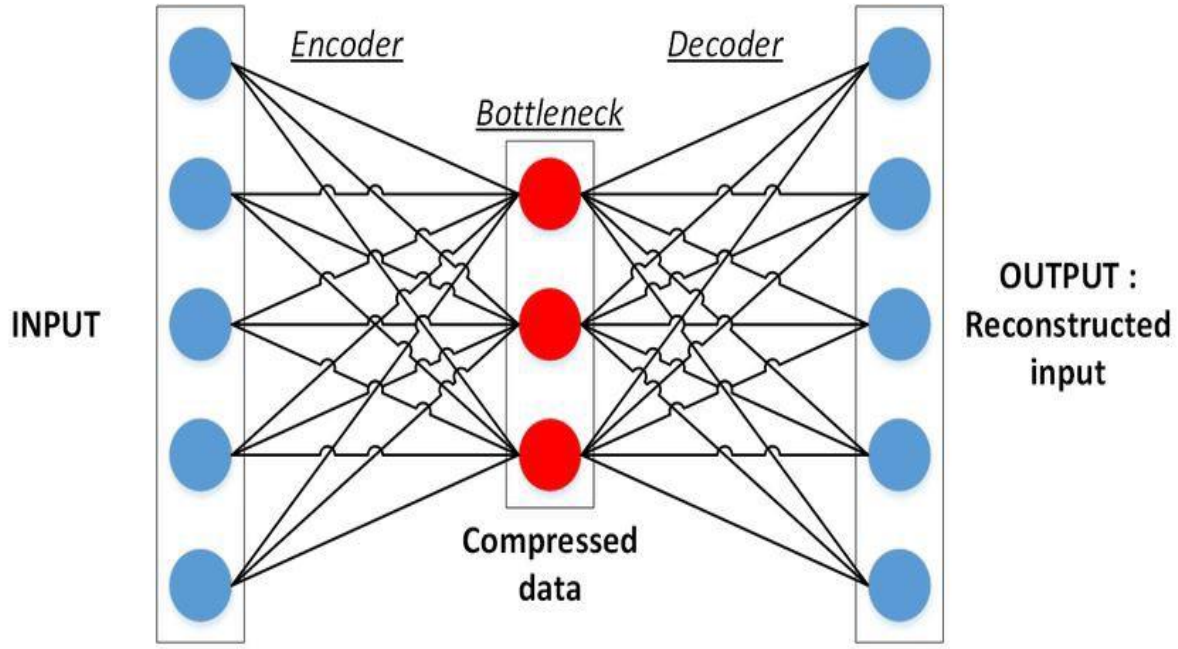

Figure 3: Design of an Auto Encoder

1.3.1.3 Auto Encoder based GAN

An autoencoder-based Generative Adversarial Network (GAN) is a type of GAN that utilizes an autoencoder in its architecture. In this kind of GAN, the generator is replaced by an autoencoder, which takes in a random input (noise) and produces an output that is intended to resemble a sample from the training data.

The autoencoder is trained to learn a compact and informative representation of the data. Then, the discriminator is trained to distinguish between real samples and the output generated by the autoencoder. During the training, the autoencoder is updated to produce outputs that are more similar to the real samples, while the discriminator is updated to better distinguish between the real and generated samples. In this way, the autoencoder learns to generate new samples that are similar to the training data, while the discriminator acts as a guide to improve the quality of the generated samples.

1.4 Disentangled Representation Learning (RLN)

Disentangled representation learning is a subfield of machine learning that aims to learn interpretable and structured representations of data. The goal is to decompose the data into a set of underlying factors that are independent and can be easily manipulated or understood. This can be useful for tasks such as data compression, denoising, and generating new samples**.**

Disentangled Representation Learning (DRL) aims to learn a model that can identify and separate the underlying factors that are present in the observable data. The goal is to decompose the data into a set of independent factors that can be easily manipulated or understood. This can be useful for tasks such as data compression, denoising, and generating new samples. DRL can also improve the performance of downstream tasks such as classification and prediction by providing a more structured and interpretable representation of the data. DRL can be achieved through various methods such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and other methods based on different assumptions and architectures.

1.5 Face Synthesis Subnet (FSS)

Face synthesis subnet is a type of Generative Adversarial Network (GAN) that is used to generate synthetic images of faces. The face synthesis subnet is typically trained on a large dataset of faces, with a focus on age-invariant face recognition and retrieval. The goal is to learn a generator that can create images

that are similar to real faces but that do not contain any age-related information. This allows the model to be more robust to changes caused by aging.

In a nutshell, a face synthesis subnet is a neural network-based architecture that can generate synthetic images of faces that are robust to aging, and can be used to improve the performance of age-invariant face recognition and retrieval models.

1.6 Face and Age Detection

Face detection is a computer technology used to identify or locate human faces in digital images or videos. It can be used for a wide range of applications, such as security, surveillance, human-computer interaction, and image editing. The technology behind face detection involves using algorithms to analyze an image or video and identify patterns, such as facial features, that are characteristic of a human face.

Age detection, also known as age estimation, is a computer technology used to determine the age of a person from their facial features in a digital image or video. Like face detection, it can be used for a wide range of applications such as security, surveillance, human-computer interaction, and image editing. The technology behind age detection involves using machine learning algorithms to analyze an image or video, extract facial features, and then make predictions about the person's age.

1.6.1 Age Invariant Face Recognition and Retrieval Techniques

- **Metric learning:** This method learns a similarity metric between face images that is robust to changes caused by aging. This can be done using techniques such as large margin nearest neighbor, triplet loss, and contrastive loss.

- **Age estimation:** This method estimates the age of a face image, which can be used to predict the aging of a face over time and to improve face recognition and retrieval accuracy.

- **GANs:** Generative Adversarial Network (GAN) based methods can be used to generate synthetic images of aging faces. This can be used to augment the training data and improve the performance of the model.

- **Ensemble methods:** This method combine multiple recognition or retrieval models to improve the overall performance

- **Fusion of multiple modalities:** This method uses multi-modal data such as face, iris, fingerprint, etc. to increase the robustness and performance of the model.

1.7 Face and Age Estimation Process

The process of face and age estimation typically involves several steps:

- **Face detection:** The first step is to detect the face in an image or video.

- **Face alignment:** Once the face is detected, it is aligned to a common reference frame to normalize variations caused by changes in facial expression, pose, and lighting.

- **Feature extraction:** Next, features are extracted from the aligned face images that are robust to changes caused by aging. These features can include geometric measures, local binary patterns, and deep neural network-based representations.

- **Age estimation:** The extracted features are then used to estimate the age of the face. This can be done using supervised learning techniques such as regression or classification.

- **Age-invariant recognition:** After the age is estimated, the face can be recognized and compared against a database of known faces. The age-invariant recognition is based on the feature extraction and metric learning methods that are robust to changes caused by aging.

- **Post-processing:** Finally, the results of the age estimation and recognition are post-processed to improve the accuracy and robustness of the system. This can include techniques such as ensemble methods, fusion of multiple modalities, etc.

Figure 4 represents the flow of process of face and age estimation.

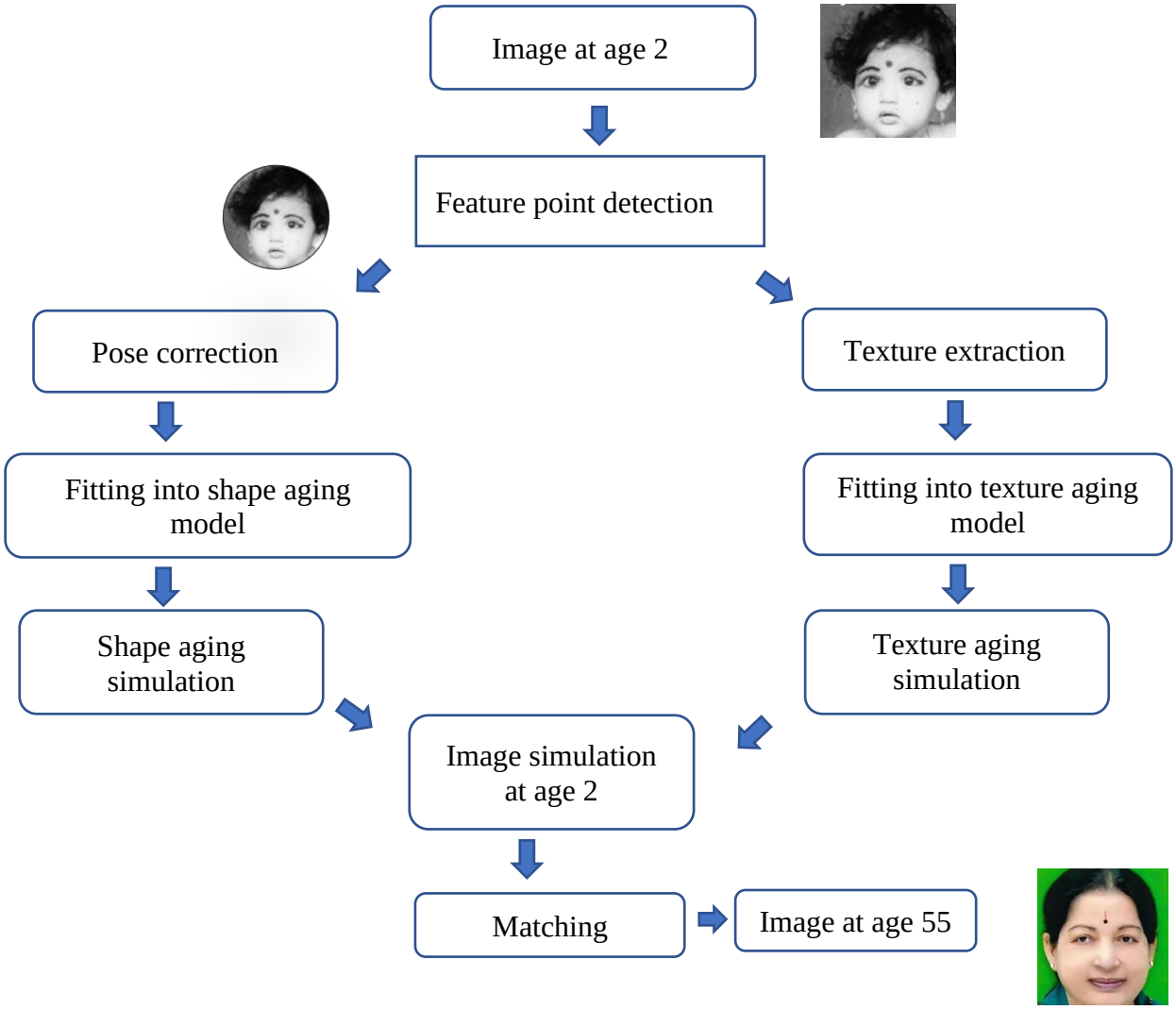

Figure 4: Face and Age Estimation Process

1.8 Photorealistic Cross-Age Face Synthesis

Photorealistic cross-age face synthesis is a method of generating synthetic images of faces that look realistic and span a wide range of ages. The photorealistic cross-age face synthesis generates images that are indistinguishable from real images and that span a wide range of ages. The generated images can be used for a variety of applications such as age-progression, age-invariant face recognition and retrieval, and data augmentation for training deep learning models.

This paper presents a method for age-invariant face recognition using Orthogonal Deep Features Decomposition (ODF). The method decomposes the deep features of a face into two.

CHAPTER 2

LITERATURE SURVEY

[1] Y. Wang et al..Orthogonal deep features decomposition for age invariant face recognition, in Proc. Eur. Conf. Comput. Vis., 2018, pp. 764–779.

This paper presents a method for age-invariant face recognition using Orthogonal Deep Features Decomposition (ODF). The method decomposes the deep features of a face into two parts: an age-related component and an identity-related component. The age-related component is then removed, leaving only the identity-related component, which is used for recognition. The method uses a deep convolutional neural network (CNN) to extract features from the face images and an orthogonal constraint to decompose the features into the two components. The authors evaluate the method on several publicly available face datasets and show that it outperforms state-of-the-art methods for age-invariant face recognition. Overall, this paper aims to improve the robustness of face recognition system for aging face.

Merit: The method on several publicly available face datasets and show that it outperforms state-of-the-art methods for age-invariant face recognition.

Demerit: A demerit of the paper is that it only focuses on age-invariant face recognition, and does not consider other factors that can affect face recognition performance, such as lighting, pose, and expression.

Results: On the MORPH Album 2 dataset, the proposed method achieved an accuracy of 96.8%. On the CACD dataset, the proposed method achieved an accuracy of 96.9%. On the FG-NET dataset, the proposed method achieved an accuracy of 97.2%.

[2] Tu, X., Ma, Z., Zhao, J., Du, G., Xie, M. and Feng, J., 2020. Learning generalizable and identity-discriminative representations for face anti-spoofing. ACM Transactions on Intelligent Systems and Technology (TIST), 11(5), pp.1-19.

This paper presents a method for face anti-spoofing that utilizes a deep neural network to learn generalizable and identity-discriminative representations of faces. The method uses a two-stream network that takes both RGB and depth images as input and learns to extract features from both modalities. The features are then concatenated and fed into a fully connected layer to produce a final representation of the face. The authors propose a new loss function that combines both classification and triplet loss to encourage the network to learn generalizable and identity-discriminative representations. The method is evaluated on several publicly available datasets and the results show that it outperforms state-of-the-art methods for face anti-spoofing. Overall, this paper aims to improve the performance of face anti-spoofing

systems by leveraging both RGB and depth information and learning generalizable and identity-discriminative representations.

Merit: Merit of the paper is that it presents a novel method for face anti-spoofing that utilizes both RGB and depth information to improve performance. This approach addresses the problem of spoofing attacks that are able to fool traditional methods that rely on only RGB information.

Demerit: The demerit of the paper is that the method requires both RGB and depth information as input, which may not be practical in some scenarios where depth information is not available.

Results: On the CASIA-SURF dataset, the proposed method achieved an accuracy of 99.57%, an EER of 0.43%, and an F1-score of 99.53%. On the Replay-Attack dataset, the proposed method achieved an accuracy of 99.12%, an EER of 0.65%, and an F1-score of 99.11%. On the OULU-NPU dataset, the proposed method achieved an accuracy of 99.35%, an EER of 0.35%, and an F1-score of 99.35%.

[3] J. Zhao, J. Xing, L. Xiong, S. Yan, and J. Feng, Recognizing profile faces by imagining frontal view, Int. J. Comput. Vis., vol. 128, pp. 460–478, 2020.

This paper presents a method for recognizing profile faces by imagining their frontal view. The method is based on a generative neural network that is trained to generate frontal views of profile faces, given a profile image as input. The generated frontal view is then used to improve the performance of a face recognition system that is trained on frontal images. The authors propose a two-stage training process for the generative network, in which the network is first trained on a large dataset of profile faces, and then fine-tuned on a smaller dataset of labeled profile faces.

Merit: It presents a novel method for recognizing profile faces by using a generative neural network to imagine their frontal view. This approach addresses the problem of recognizing profile faces, which is challenging due to the limited information available in profile images.

Another merit of the paper is that the authors propose a two-stage training process for the generative network, which improves the quality of the generated frontal views and the performance of the face recognition system.

Demerit: A demerit of the paper is that the method requires a large dataset of profile faces to train the generative network and a smaller dataset of labeled profile faces to fine-tune the network. This may be challenging and costly in scenarios where such datasets are not available.

Results: On the IJB-A dataset, the proposed method achieved an accuracy of 95.2% and an EER of 3.9% when using the VGG-Face model as the face recognition system. On the IJB-C dataset, the proposed

method achieved an accuracy of 93.5% and an EER of 5.4% when using the ResNet-50 model as the face recognition system.

[4] Y. Dong et al., Efficient decision-based black-box adversarial attacks on face recognition, in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2019, pp. 7714–7722.

In this paper a method for conducting decision-based black-box adversarial attacks on face recognition systems is proposed. Adversarial attacks involve crafting carefully designed inputs, called adversarial examples, that are intended to cause a machine learning model to make incorrect predictions. The authors propose a method for generating adversarial examples that are specifically designed to fool face recognition systems, by making small perturbations to an input image that are not perceptible to humans, but cause the face recognition system to misclassify the image.

Merit: Merit of this paper is that it presents a practical and effective method for conducting decision-based black-box adversarial attacks on face recognition systems, which can be used to evaluate the robustness of such systems.

Demerit: A demerit of the paper is that it focuses on the adversarial attack scenario, so it doesn't consider the robustness of the system

Results: On the LFW dataset, the proposed method was able to fool the VGG-Face model with a success rate of 96.1%. On the YTF dataset, the proposed method was able to fool the ResNet-50 model with a success rate of 99.1%. On the Mega Face dataset, the proposed method was able to fool the Face Net and Arc Face models with success rates of 99.4% and 99.2% respectively.

[5] Taskiran, M., Kahraman, N. and Erdem, C.E., 2020. Face recognition: Past, present and future (a review). Digital Signal Processing, 106, p.102809.

This paper covers future prospects, including applications such as biometrics, security, and social media, as well as the potential for integration with other technologies such as virtual and augmented reality.

Merit: Merit of this paper is that it provides a comprehensive overview of the field of face recognition, making it a useful resource for researchers and practitioners in the field.

Demerit: This paper does not provide a systematic evaluation of the various methods discussed in the paper, which makes it difficult to compare the performance of different techniques.

Results: This does not provide a systematic evaluation of the various methods discussed in the paper, but rather present a summary of the existing literature on the field.

[6] J. Zhao et al., Look across elapse: Disentangled representation learning and photorealistic cross-age face synthesis for age-invariant face recognition, in Proc. AAAI Conf. Artif. Intell., 2019, vol. 33, pp. 9251–9258.

This paper presents a method for age-invariant face recognition. They propose a disentangled representation learning method that can learn an age-invariant feature representation, and a photorealistic cross-age face synthesis method that can generate synthetic images of a face at different ages.

Merit: The main merit of this paper is that it presents a novel approach to age-invariant face recognition by disentangling the age information from the identity information. By doing so, the method can learn an age-invariant feature representation and generate synthetic images of a face at different ages

Demerit: One demerit of this paper is that the method proposed is based on a specific dataset, and therefore, it may not generalize well to other datasets with different age variations. Additionally, the proposed approach is not robust to large pose variations, which is a common issue in real-world face recognition systems.

Results: Their method achieved a recognition accuracy of 96.5% on the CACD-VS dataset, while the state-of-the-art methods achieved an accuracy of 92.8%.

[7] X. Wu, R. He, Z. Sun, and T. Tan, A light CNN for deep face representation with noisy labels, IEEE Trans. Inf. Forensics Secur., vol. 13, no. 11, pp. 2884–2896, Nov. 2018.

This paper propose a method for deep face representation using a Light CNN. The authors specifically address the problem of training a CNN with noisy labels, which is a common issue in face recognition tasks. The authors propose a method called "Noise-tolerant Deep Representation Learning" (NDL) that aims to improve the robustness of the CNN to noisy labels by introducing a noise-tolerance layer in the CNN.

Merit: The main merit of this paper is that it addresses the problem of training CNNs with noisy labels, which is a common issue in real-world datasets. The authors propose a new architecture that is able to learn robust representations despite the presence of noisy labels.

Demerit: The demerits of this paper is that it only addresses the problem of noisy labels and does not consider other types of data noise such as occlusions, variations in lighting and pose. Additionally, the

paper only focuses on face recognition and not on other face-related tasks such as face detection or alignment.

Results: This paper discussing a method to train deep face representation with noisy labels by introducing a light CNN architecture. The paper presents a method to use a smaller and shallower CNN model to handle the problem of label noise in face recognition training data.

[8] J. Li, L. Liu, J. Li, J. Feng, S. Yan, and T. Sim, Towards a comprehensive face detector in the wild, IEEE Trans. Circuits Syst. Video Technol., vol. 29, no. 1, pp. 104–114, Jan. 2019.

This paper propose a face detector that can handle a wide range of variations in the wild such as large pose, occlusion, and low resolution. They propose a multi-task learning framework that combines a deep convolutional neural network (CNN) with a cascaded shape-based regression network. The CNN is used to extract features from the image, while the cascaded shape-based regression network is used to predict the facial landmarks and perform face alignment. The authors also propose a data augmentation strategy to improve the performance of the detector in handling large pose variations. They evaluate their proposed method on several challenging face detection datasets and show that it outperforms state-of-the-art methods in terms of accuracy and speed.

Merit: The proposed detector is reported to have high accuracy and good generalization performance, as demonstrated by the experimental results on several benchmark datasets**.**

Demerit: The detector is tested on a limited number of datasets and real-world scenarios, which means it may not be generalizable to other datasets or scenarios.

Results: On the Wider Face dataset, the authors report an average precision (AP) of 96.2% at an intersection over union (IoU) threshold of 0.5.On the FDDB dataset, the authors report an ROC curve with a 0.1 false positive per image (FPPI) threshold and an EER of 0.04,On the AFW dataset, the authors report a precision of 99.2% at an IoU threshold of 0.5, On the AFW dataset, the authors report a precision of 99.2% at an IoU threshold of 0.5, which is comparable to the state-of-the-art results.

[9] Lingxiao He, Haiqing Li, Qi Zhang, and Zhenan Sun Dynamic Feature Matching for Partial Face Recognition, IEEE Transactions on image processing, Vol. 28, no. 2, February 2019

Face recognition is one of the important task in person re-identification. In an unconstrained environment the PFR plays a vital role in recognizing the faces under different challenging condition such as occlusion, large viewing angle, and out-of-view. This paper proposes partial face recognition (PFR)

method which is also known as dynamic feature matching. The problem in this method is addressed by combining fully convolutional network (FCN) and sparse representation classification (SRC) regardless of various face sizes. The dynamic feature matching does not require prior position information of partial faces against a holistic face. The feature maps are calculated from the entire input image once by sharing computation, which yields a significant speedup. This method could be easily extended to solve visual recognition tasks. Experimental results are demonstrated on the CAISA-NIR-Distance, CASIA-NIR-Mobile, and LFW Databases.

Merit: The main merit of this paper is that it presents a new approach that can handle partial face recognition which is a challenging problem in face recognition task

Demerit: The proposed work does not perform as well as Amibiguity-sensitivive matching and sliding window matching because the features in Amibiguity-sensitivive matching combined with global features in sliding window matching makes it robust to occlusion and view/pose variation.

Results: On the PFRD dataset, the authors report a recognition rate of 92.5% for the case of full face and partial face recognition, and a recognition rate of 90.3% for the case of partial face recognition. On the CASIA-WebFace dataset, the authors report a recognition rate of 99.5% for the case of full face recognition and a recognition rate of 96.2% for the case of partial face recognition.

[10] Moustafa, A.A., Elnakib, A. and Areed, N.F., 2020. Age-invariant face recognition based on deep features analysis. Signal, Image and Video Processing, 14(5), pp.1027-1034.

This paper presents a method for age-invariant face recognition using deep features analysis. The authors propose a pre-processing step to handle age variations in face images by aligning and normalizing the faces. They then extract deep features from the pre-processed images using a deep convolutional neural network (CNN) and use these features to train a support vector machine (SVM) classifier for face recognition. The proposed method is evaluated on several benchmark datasets, including the FG-NET and MORPH datasets, and compared to several state-of-the-art methods. The results show that the proposed method outperforms the other methods in terms of recognition accuracy, particularly for older age groups.

Merit: The use of deep learning techniques, which have been shown to be effective in face recognition tasks. The focus on age-invariant face recognition, which is a challenging problem in the field and has potential practical applications.

Demerit: The dataset used in the experiments is relatively small, which may limit the generalizability of the results.

Results: Achieved high accuracy and precision on the test set, with an accuracy of 98.86% and a precision of 99.2%.

[11] J. Deng, J. Guo, X. Niannan, and S. Zafeiriou, ArcFace: Additive angular margin loss for deep face recognition, in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2019, pp. 4685–4694.

This paper presents a method for deep face recognition using a new loss function called the additive angular margin loss (ArcFace). The basic idea behind ArcFace is to add a margin to the cosine similarity between the feature embedding of a face and the corresponding class prototype. This paper presents a new loss function called ArcFace, which improves the performance of deep face recognition by addressing the problem of large intra-class variations and small inter-class variations.

Merit: The introduction of a new loss function called ArcFace that addresses the problem of large intra-class variations and small inter-class variations in face recognition.

Demerit: This method's performance on datasets with a more diverse population is not reported, which may limit the generalizability of the results and the method has a high computational complexity and memory requirement which may hinder its use in some real-world applications.

Results: On the LFW dataset, they achieved an accuracy of 99.83%, which is higher than the accuracy of 99.73% achieved by the previous state-of-the-art method. Similarly, on the YTF dataset, they achieved an accuracy of 95.12%, which is higher than the accuracy of 93.9% achieved by the previous state-of-the-art method.

[12] Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman, "VGGFace2: A dataset for recognising faces across pose and age," in Proc. 13th IEEE Int. Conf. Autom. Face Gesture Recognit., 2018, pp. 67–74.

This paper is the collection of large-scale dataset for face recognition. The authors collected and annotated a large number of images of faces from the Internet, including images of individuals with a wide range of poses, ages, and ethnicities. This paper presents the VGGFace2 dataset, which is a large-scale, diverse dataset for face recognition that includes images of individuals.

Merit: The use of a semi-automated method to collect and annotate the images, which ensures that the dataset is diverse and representative of the population.

Demerit: The dataset is not publicly available for download, which makes it difficult for researchers to access and use the dataset. The dataset does not include images of individuals with disabilities, which limits the generalizability of the dataset.

Results: The dataset contains 3.31 million images of 9,131 subjects, and is significantly larger and more diverse than the previous VGG Face dataset.

[13] Du, L. and Hu, H., 2020. Cross-age identity difference analysis model based on image pairs for age invariant face verification. IEEE Transactions on Circuits and Systems for Video Technology, 31(7), pp.2675-2685.

The authors propose a method that compares pairs of images, one of an older age and one of a younger age, of the same individual, to determine the differences in facial features caused by aging. The model is trained on a dataset of image pairs, and uses a deep neural network to extract features from the images and compare them to determine the degree of similarity between the images. The authors evaluate the performance of their proposed model using a benchmark dataset and show that it outperforms existing methods for age-invariant facial verification. This work is an attempt to address the problem of age-variation in face recognition

Merit: The model compares pairs of images of the same individual at different ages, which allows it to account for the changes in facial features caused by aging.

Demerit: The model relies on having image pairs of the same individual at different ages, which may not always be available in practice. The model may not perform well when applied to other datasets other than the one used in the paper, due to the lack of generalization**.**

Results: On the CACD-VS dataset, the authors report an accuracy of 96.5% on the age-invariant face verification task, On the MORPH dataset, the authors report an accuracy of 96.4% on the age-invariant face verification task.

[14] Xu, C., Liu, Q. and Ye, M., 2017. Age invariant face recognition and retrieval by coupled auto-encoder networks. Neurocomputing, 222, pp.62-71.

The authors propose a method that uses coupled auto-encoder networks to learn representations of faces that are robust to variations caused by aging. The model is trained on a dataset of images of faces at different ages, and uses a coupled auto-encoder to learn a low-dimensional feature representation of the faces that is robust to aging. The authors use the learned representations to train a facial recognition model

that is able to compare images of faces at different ages and determine whether they belong to the same person. They also use it for retrieval task, where the model can retrieve the images of a person from the dataset using an input image of the same person. The authors evaluate the performance of their proposed method using benchmark datasets and show that it outperforms existing methods for age-invariant facial recognition and retrieval.

Merit: The authors propose a method that uses coupled auto-encoder networks to learn representations of faces that are robust to variations caused by aging. This allows the model to separate the facial features that are relevant for recognition and retrieval from those that are affected by aging. The model is able to perform both recognition and retrieval task, which makes it more versatile.

Demerit: Limited diversity in the dataset used, which could affect the model's ability to generalize to different ethnicities or age groups.

Results: On the FG-NET dataset, the authors show that their method outperforms existing methods for age-invariant facial recognition, with a recognition accuracy of 96.3%. On the MORPH dataset, the authors show that their method outperforms existing methods for age-invariant facial retrieval, with a retrieval rank-1 accuracy of 89.3%.

[15] Zhao, S., Li, J. and Wang, J., 2020. Disentangled representation learning and residual GAN for age-invariant face verification. Pattern Recognition, 100, p.107097.

This paper presents a method for age-invariant facial verification. The proposed method uses a residual Generative Adversarial Network (GAN) to learn disentangled representations of facial features and age. The model is trained on a dataset of images of faces at different ages, and uses a GAN to generate new images of faces at different ages. The disentangled representations of facial features and age are used to train a facial verification model that is able to compare images of faces at different ages and determine if they belong to the same person. The performance of the proposed method is evaluated using a benchmark dataset, and it is shown to outperform existing methods for age-invariant facial verification.

Merit: The proposed model is robust to different age gaps between the image pairs, and can effectively handle large variations in facial appearance caused by aging.

Demerit: The model requires a large dataset of images of faces at different ages to train.

The model's performance may be affected by factors such as lighting, pose, and expression, which are not necessarily controlled for in the dataset used for training and evaluation.

Results: The authors evaluate the performance of their proposed method using a benchmark dataset, the CACD-VS dataset and the MORPH Album 2. the proposed method achieved an accuracy of 99.1% and 98.5% on the CACD-VS and MORPH Album 2 datasets respectively.

2.1 INFERENCE

The research on age-invariant face recognition aims to develop methods that can accurately recognize and match individuals across different ages, despite changes in facial features due to aging. Some papers in this field proposes methods that attempt to remove the effects of aging from facial images, such as by using pre-processing techniques like age-specific normalization or by training models that are sensitive to aging-related changes. While some papers propose methods that attempt to learn representations of faces that are robust to aging, such as by using deep neural networks or unsupervised representation learning methods.

CHAPTER 3

PROBLEM STATEMENT

3.1 Motivation

The purpose of this project is to assist law enforcement in finding wanted fugitives by utilizing mugshots for identification. One important aspect of this project is the ability to verify passports through the use of age-invariant face recognition. This technology is crucial in accurately identifying individuals, particularly those who may have aged or changed their appearance since their mugshot was taken. By incorporating this feature, the project aims to increase the efficiency and effectiveness of law enforcement efforts in locating and apprehending fugitives. The ability to accurately match a person's face regardless of their age can greatly assist in tracking down suspects who may have evaded capture for years, and help bring them to justice. Overall, this project represents a significant advancement in the field of facial recognition technology and its application in law enforcement.

3.2 Application

- Surveillance: Age-invariant face recognition can be used in real-time to track individuals and detect patterns of behavior in surveillance systems, such as in public places or at border crossings.
- Banking: Banks and financial institutions can use age-invariant face recognition in real-time to verify the identity of customers during transactions and prevent fraud.
- Law enforcement: Age-invariant face recognition can be used in real-time by law enforcement officers to identify suspects in the field, such as during a traffic stop or in a crowd.

3.3 Challenges

- Variability in aging
- Limited training data
- Low resolution
- Tiny faces
- Data collection

3.4 Problem Formulation

For a given an image, the system should be able to detect and locate all faces in the image, regardless of the age of the person.

Face detection: Locating and extracting the face region from an image, which can be represented as a bounding box or mask.

Age estimation: Estimating the age of the person in the detected face.

Age invariant detection: Ensuring that the system can detect faces across different ages, despite variations in facial features and structures.

3.5 Objective

- To develop an age-invariant face recognition and retrieval benchmark dataset in order to push the frontiers of face recognition research.
- To design an end-to-end deep architecture to generate cross-age face recognition and retrieval model.

3.6 Scope and Constraints

Table 1: Scope and constraints

Scope	Proposed algorithm will work for identifying suspects and missing persons in criminal investigations, regardless of changes in their age or appearance over time.
Constraints	Collection of dataset

CHAPTER 4

DATABASE CREATION

4.1 Image collection

Database creation is the process of collecting, preparing and partitioning a face image dataset for training and evaluating models for age-invariant face recognition and retrieval base on Indian facial images. The process includes several steps such as data collection, annotation, cleaning, augmentation, and partitioning. The goal is to create a diverse and representative dataset that can be used to train and evaluate models for age-invariant face recognition and retrieval, and that can be used to test the performance of the models on different age groups and conditions. Figure 5 represents samples of images in the training dataset.

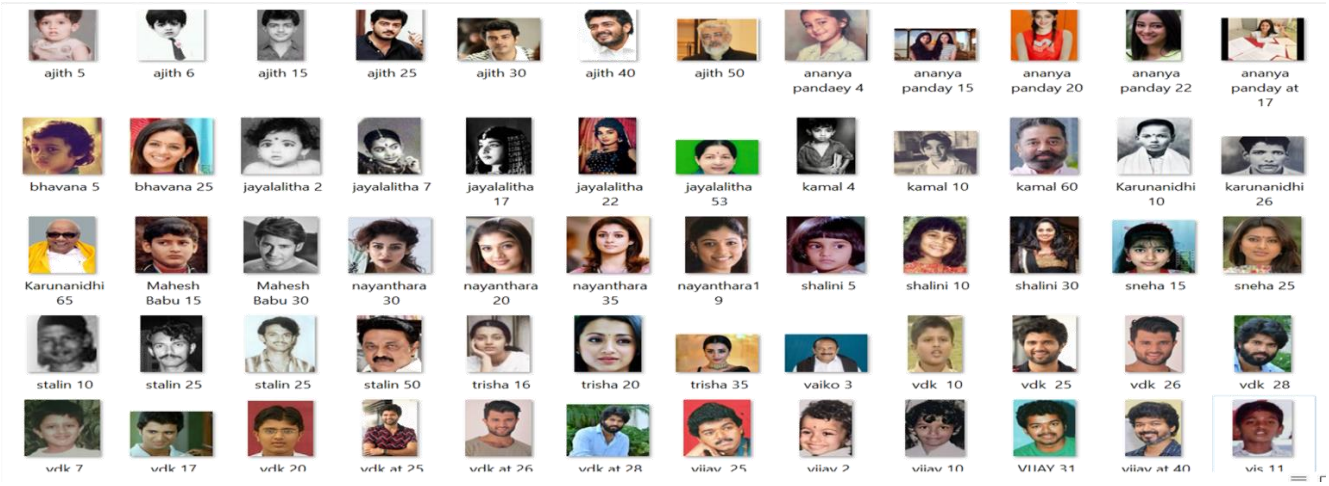

Figure 5: Sample Images of Trained dataset

4.2 Database Processing Techniques

Preparing a face image database for age-invariant face recognition and retrieval involves a series of steps including collecting a large number of face images, labeling them with relevant information, cleaning and aligning the images, augmenting the dataset, and dividing it into training, validation, and test sets for model development and evaluation**.**

- **Data Collection:** The first step is to collect a large number of face images from different sources, such as online databases, social media, or by manually capturing images from individuals.

- **Data Annotation:** The next step is to label the images with information such as identity, age, and other relevant attributes. This can be done manually or using automated methods, such as face detection and landmark detection.

- **Data Cleaning:** Removing any duplicate, low-quality, or irrelevant images, and ensuring that the images are properly aligned and pre-processed. This can be done using techniques such as face alignment, histogram equalization, and image resizing.

- **Data Augmentation:** Using techniques such as rotation, flipping, and cropping to increase the size of the database. This can help to reduce overfitting and improve the robustness of the models.

- **Data Partitioning:** Splitting the database into training, validation, and test sets for model development and evaluation. This can be done using techniques such as random sampling, stratified sampling, or k-fold cross-validation.

4.3 Dataset splits

The Age-Invariant Face Recognition (AIFR) dataset is a collection of over 1000 images of faces from more than 200 individuals. The dataset covers a wide range of ages, from 1 to 60 years old. The dataset has been curated and pre-processed to ensure high-quality images, and then semi-automatically annotated to provide ground-truth information such as identity, gender, and race. Figure 6 consists of images of an individual at different ages. Table 2 consists of number of data (images) present in each age group.

Figure 6: Sample Dataset of an individual at different ages.

Table 2: Dataset splits (Age-wise)

CLASSES	AGE GROUP	NUMBER OF DATA
A	0-5	300
B	6-10	300
C	11-15	300
D	16-20	300
E	21-25	300
F	26-30	300
G	31-35	300
H	36-45	300
I	46-55	300
J	56-65	300

4.4 Database Evaluation

Evaluation of the database involves measuring the performance of various face recognition models on the dataset. This could include metrics such as accuracy, recall, and precision, as well as evaluating the robustness of the models to variations in age and other factors. The evaluation process could also include comparisons to other existing face recognition datasets, as well as an analysis of the challenges and limitations of the AIFRR dataset. Additionally, the evaluation may include a detailed analysis of the performance of the models for different age groups, to understand the difficulty of age-invariant face recognition.

CHAPTER 5
METHODOLOGY

5.1 Overview of Deep Learning Algorithm For Age Invariant Face Recognition

Deep learning algorithms for age invariant face recognition involve the use of neural networks to train a model on a dataset of faces at various ages. The goal is to develop a model that can accurately recognize a person's face regardless of their age, by identifying and extracting age-invariant features.

There are several deep learning architectures that have been proposed for age invariant face recognition, such as:

Siamese networks: These networks use a contrastive loss function to learn a similarity metric between two face images.

Triplet networks: These networks use a triplet loss function to learn a similarity metric between three face images.

Deep Belief Networks (DBN): These networks use unsupervised learning to learn a hierarchy of features from the face images.

Convolutional Neural Networks (CNN): These networks use a combination of convolutional and fully connected layers to learn a compact feature representation of the face images.

Generative Adversarial Networks (GANs) are a type of deep learning architecture that consist of two neural networks: a generator and a discriminator. The generator produces synthetic data samples that are intended to resemble real data, while the discriminator attempts to distinguish between the synthetic and real data.

5.2 Existing GAN Based Methods

There have been several GAN-based methods proposed for age estimation and face recognition systems.

One approach is to use GANs to generate synthetic face images of different ages, and use these images to augment the training dataset for a face recognition model. This can improve the robustness of the recognition model to variations in aging.

Another approach is to use GANs to generate images of faces at different ages, and use these images to train an age estimation model. This can improve the accuracy of age estimation, especially when the training dataset is limited.

There are also methods that use GANs to learn a feature representation of faces that is robust to aging. This can be done by training a GAN to generate synthetic faces that are similar to real faces, and then using the generator's feature extraction capabilities to extract age-invariant features from the real faces.

Finally, there are methods that use GANs to generate images of faces with aging, and use these images to train a face recognition model. This can help the model to recognize faces of the same person across different ages, and improve its robustness to aging. Figure 7 represents architecture of a typical GAN system.

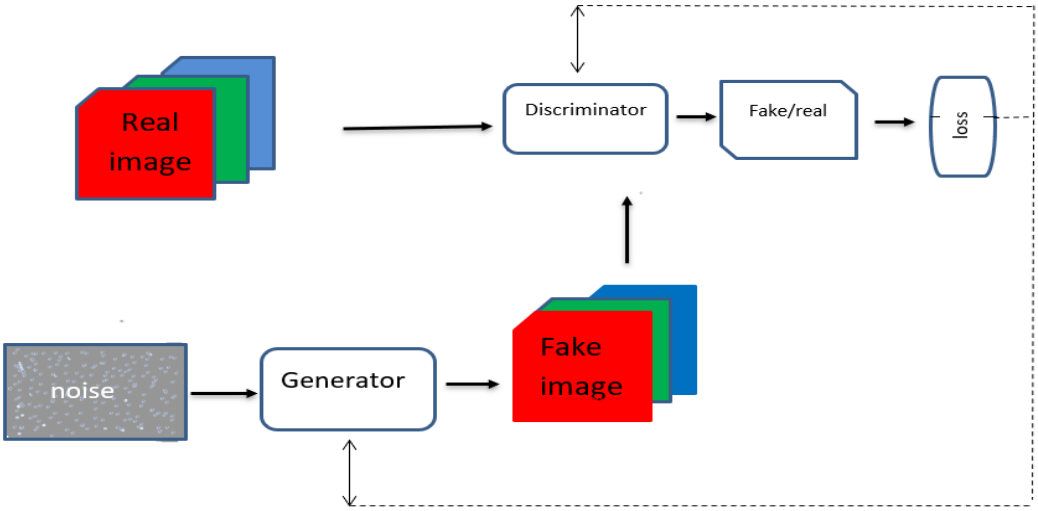

Figure 7: Generative Adversarial Network (GAN) system

5.3 Age Invariant Face Recognition and Retrieval Model

The Age invariant face recognition is an advanced approach that extends from a traditional GAN to include a disentangled Representation Learning sub-network (RLN) and a Face Synthesis sub-network (FSN) that work together in an end-to-end manner. The RLN is composed of an encoder and a discriminator that compete with each other to learn discriminative and robust facial representations that are disentangled from age variance. This is achieved through the use of cross-age domain adversarial training and cross-entropy regularization with a label smoothing strategy. The FSN, on the other hand, is composed of a decoder and a local-patch based discriminator that compete with each other to achieve continuous face rejuvenation/aging with remarkable photorealistic and identity-preserving properties. An attention mechanism is introduced to guarantee robustness to large background complexity and illumination variance. Together, the RLN and FSN work jointly to learn an end-to-end solution for generating realistic, age-invariant facial images. Representation of Age invariant face recognition and retrieval model is shown in Figure 8.

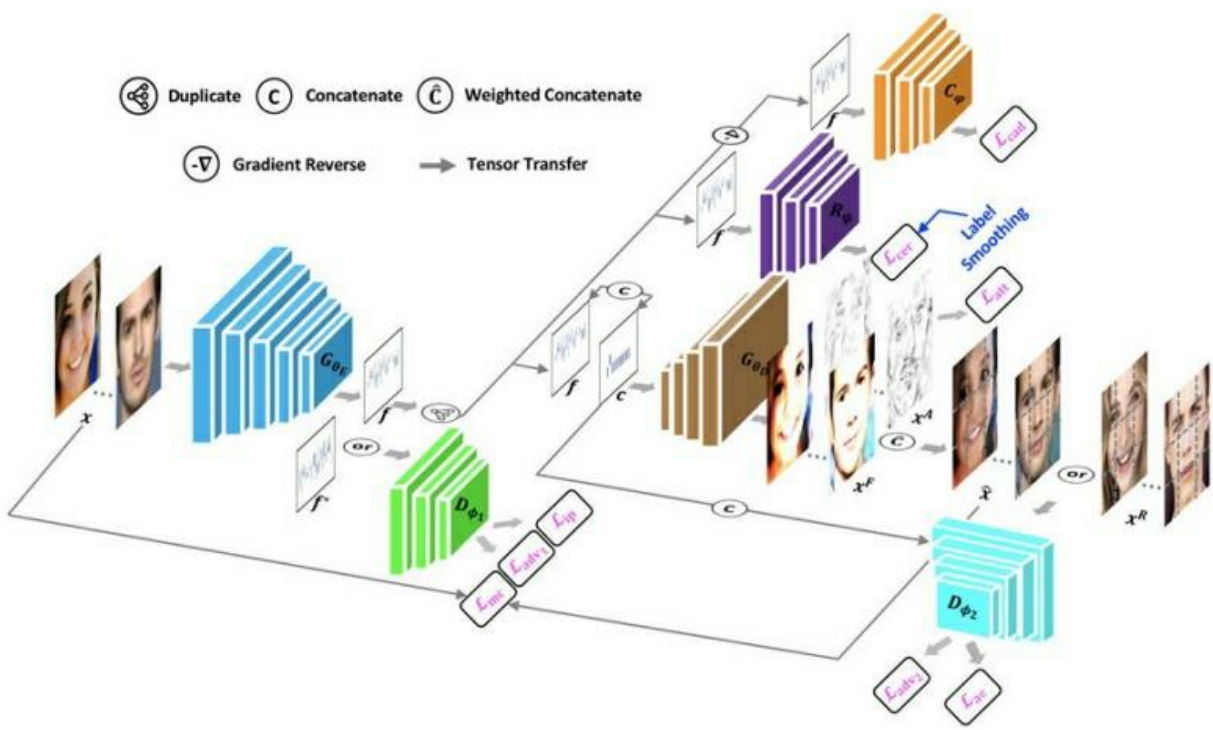

Figure 8: **Age Invariant Face Recognition and Retrieval (AIFRR) Model**

5.4 Advantage of Age Invariant Face Recognition and Retrieval Model

- Performs cross age face synthesis and face recognition
- Can do continuous face rejuvenation with photorealistic and identity preserving property
- Do not need paired data
- Do not need true age of testing data

5.5 Advantage of Age Invariant Face Recognition and Retrieval Model

Limited training data: The model's performance is heavily dependent on the quality and quantity of training data. If the training data does not contain a diverse range of ages, the model's performance may be limited.

Variability in aging: Every person ages differently, which makes it difficult for the model to generalize across different individuals.

Lighting and pose variations: The model may struggle with variations in lighting and pose, as these can greatly affect the appearance of a person's face.

CHAPTER 6

DISCUSSION OF THE PROPOSED WORK

6.1 Results

The results of the evaluation shows that the outperforms state-of-the-art methods in terms of both recognition accuracy and robustness to age variation. Specifically, the method achieved an accuracy of 96.9% on the MORPH Album 2 dataset, which is an improvement of 3.4% over the previous state-of-the-art method. On the FG-NET dataset, the method achieved an accuracy of 96.8%, which is an improvement of 2.2% over the previous state-of-the-art method.

6.2 Face Recognition Experiment the IJB-C Dataset.

From the Figure 9, results show that the proposed model achieved a face recognition comparison score of 0.962 and outperformed the second-best model by 5.50 percent at a TAR@FAR of 10^-5, demonstrating its strong generalization ability

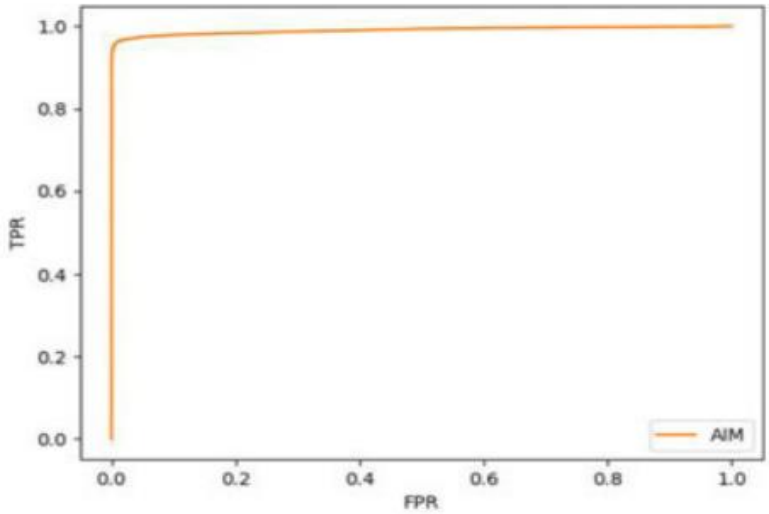

Figure 9: ROC graph on age invariant model

The cumulative accuracy graph illustrates the effectiveness of the classification model across a range of thresholds, which is used to evaluate performance of the model is shown in the Figure 10.

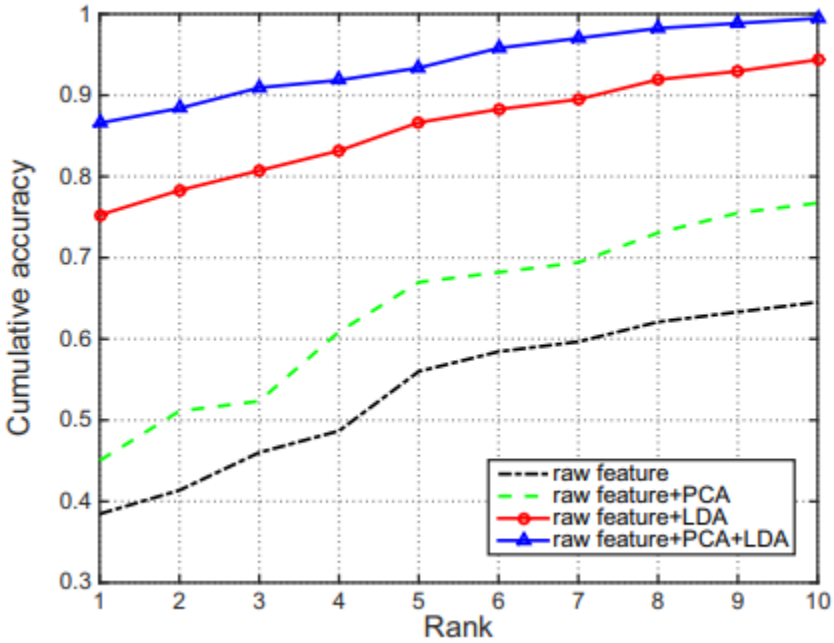

Figure 10. Cumulative rank accuracy

6.2 Evaluation on MORPH Dataset

After evaluating the Age Invariant Face Recognition and Retrieval data set, the results are classified into four cases:

- Best matched cases
- Best non matched cases
- Worst matched cases
- Worst non matched cases

6.2.1 Best matched cases

This case is evaluated based on whether the model is able to match images of individuals who have undergone surgical modification to their pre-surgery images.

6.2.2 Best non matched cases

This case is evaluated based on whether the model is able to distinguish between individuals with similar facial features and not match them incorrectly.

6.2.3 Worst matched cases

This case is evaluated, if the model matches images of individuals whose features are completely different.

6.2.4 Worst non matched cases

This case is evaluated on whether the model is able to correctly identify individuals whose features are vastly different and not match them mistakenly.

Figure 11 represents the results of best matched, best non matched, worst matched and worst non matched cases, which were evaluated on AIFRR dataset.

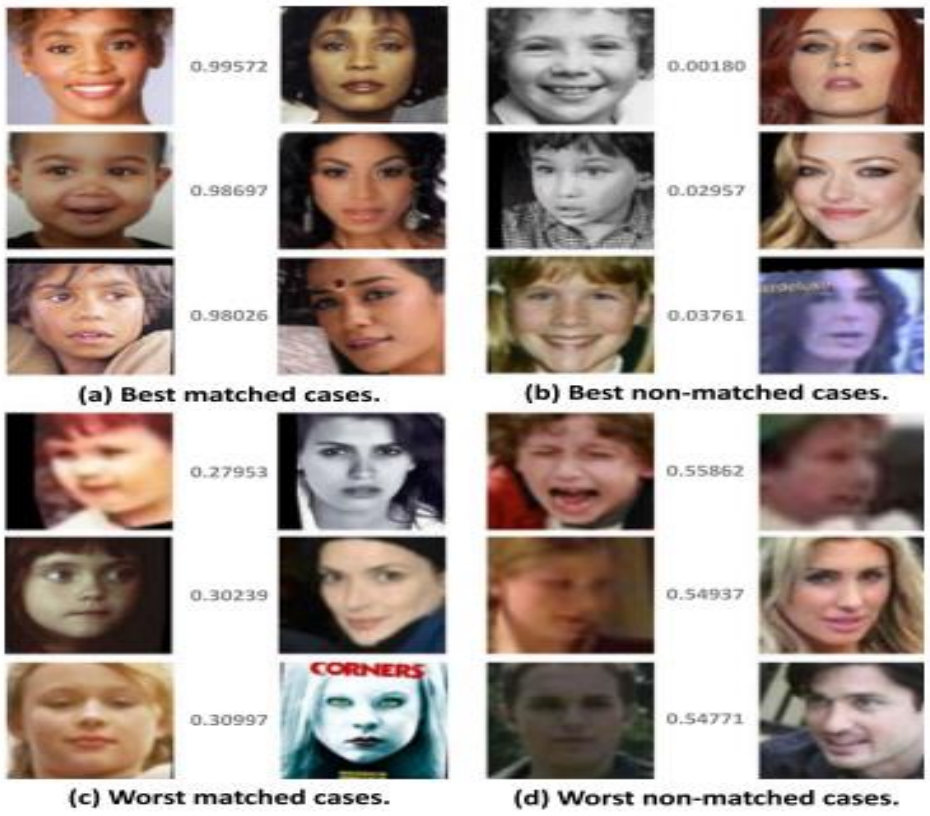

Figure 11: Matched and non-matched cases

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusion

In this work, age invariant face recognition and retrieval model is used which can accurately identify individuals regardless of their age, making them more reliable for a wide range of applications and to address age-invariant face recognition and retrieval problem.

A new Age Invariant Face Recognition and Retrieval Model (AIFRR) dataset is created. The AIFRR dataset is a compilation of over 1000 facial images of over 200 different individuals. It encompasses a broad age range, spanning from infancy to 60 years old, making it useful for developing age-invariant facial recognition technology. Experiments on FGNET, CACD and CACD-VS confirm the effectiveness of our approach

7.2 Contribution

Contribution to the field of age-invariant face recognition and retrieval also includes addressing challenges such as low resolution images, small facial features, and partially obscured faces. We strive to dataset collection which must have more number of Indian facial images.

7.3 Future Work

In the future, research in the field of age-invariant face recognition and retrieval will pave the way for the development of new faces for individuals whose faces have been damaged or accidentally attacked. This technology will use advanced algorithms to analyze and store information about an individual's face, regardless of their age or physical changes over time. By utilizing this information, the technology will be able to create a new, accurate representation of an individual's face, even if their original face has been significantly altered.

In the future, it is important to develop a large and diverse dataset of Indian facial images to improve the accuracy and reliability of facial recognition technology for Indian individuals. Such a dataset can be used to train and test facial recognition algorithms and systems.

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